MILITARY SPECIFICATION

INDICATOR, TORQUEMETER, AEU-13/A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of engine torquemeter indicator for turboprop engines.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

DD-G-451	- Glass, Plate, Sheet, Figured (Float, Flat, Glazing, Corrugated, Mirrors and Other Uses)
QQ-S-763	- Steel Bars, Shapes, and Forgings-Corrosion Resisting
PPP-B-601	- Box, Wood, Cleated Plywood
PPP-B-636	- Box, Shipping, Fiberboard

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-C-675	- Coating of Glass Optical Elements (Anti-Reflection)
MIL-E-5400	- Electronic Equipment, Airborne, General Specification for
MIL-C-5541	- Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-7742 - Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-B-27497 - Bearings, Jewel, Sapphire or Ruby, Synthetic

STANDARDS

MILITARY

MIL-STD-100 - Engineering Drawing Practices
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-143 - Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704 - Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781 - Reliability Tests, Exponential Distribution
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810 - Environmental Test Methods
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS21209 - Insert, Screw Thread, Coarse and Fine, Screw Locking, Helical Coil, Crest
MS33558 - Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33639B - Case, Instrument, Clamp Mounted; Aircraft
MS33678 - Connector, Receptacle, Electric, Integral Mounting

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HANDBOOKS

MILITARY

MIL-HDBK-275 - Guide for Selection of Lubricants, Fluids and Compounds for use in Flight Vehicles and Components

PUBLICATIONS

STANDARDIZATION DOCUMENT

SD-6 - Provisions Governing Qualification

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. Indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.6). Qualification must be accomplished within the provisions and requirements of SD-6. That document requires that tests shall be monitored by a Government representative designated by the preparing activity or its authorized agent. The test procedure shall be in accordance with this specification. All tests required by the specification shall be successfully accomplished and documented. Certification of test results is required by the witnessing Government representative.

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials used in the manufacture of indicators shall be suitable for the purpose and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used provided such specifications are approved by the Government and contain provisions for adequate tests. The use of the contractor's specification will not constitute waiver of Government inspection.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity.

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However, if they are used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.2 Metals. Metals shall be corrosion-resistant or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in initmate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.4 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicators except where magnetic materials are essential.

3.3.5 Corrosive fumes. Materials used in the construction of the indicators shall not produce corrosive or deleterious fumes under the test conditions specified herein.

3.3.6 Protective treatment. When materials are used in the construction of the indicators that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction.

3.4.1 General. The indicator assembly shall be designed to measure the displacement between the pulse signals generated by an electromagnetic torquemeter pickup assembly, and to convert this displacement to provide indication of engine torque.

3.4.2 Construction. The indicator shall be so constructed that nc parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service. Pivots, bearings, and gears shall not bind nor shake and shall be as nearly frictionless as possible.

3.4.3 Case. The case shall conform to figure 1 and shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished

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without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon material which will be affected by the action of any atmosphere to which the indicator may be subjected.

3.4.3.1 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be either a mixture of 88 to 92 percent nitrogen and the remainder helium or 100 percent helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi.

3.4.3.2 Sealing. When a pressure of approximately zero psia is applied to the exterior of the indicator case, the leak rate shall not exceed 0.10 micron cubic foot per hour when 10 percent helium is used in the filling medium and shall not exceed 1.0 micron cubic foot per hour when 100 percent helium is used as the filling medium.

3.4.4 Coverglass. The coverglass shall conform to DD-G-451 (Type I, Class I, Quality q1) and be clear, flat, and free from flaws that would interfere with the normal reading of the instrument. The thickness of the coverglass shall be of the minimum practicable and entirely suitable for the purpose.
NOTE: Glass in accordance with the above specification, type, and quality which has had an anti-reflection coating applied in accordance with MIL-C-675 is acceptable.

3.4.4.1 Case to coverglass distance. The distance between the outer surface of the coverglass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

3.4.4.2 Dial to coverglass distance. The distance between the indicator dial and the inside of the coverglass shall be the minimum practicable and shall not exceed 0.188 inch.

3.4.5 Visibility of dial. The pointer, numerals, at least 1/3 of the shortest graduations in the range from -4,000 to 27,000 in -lb and all other specified markings on the dial shall be visible from any point within the frustum of a cone whose side makes an angle of 30° with a perpendicular to the dial and whose small diameter is the aperture of the case.

3.4.6 Dial marking. The dial shall be marked as shown in figure 2. Unless otherwise specified, all visible portions of the dial shall be lusterless black color no. 37038 of FED-STD-595 and all markings on the dial shall be lusterless

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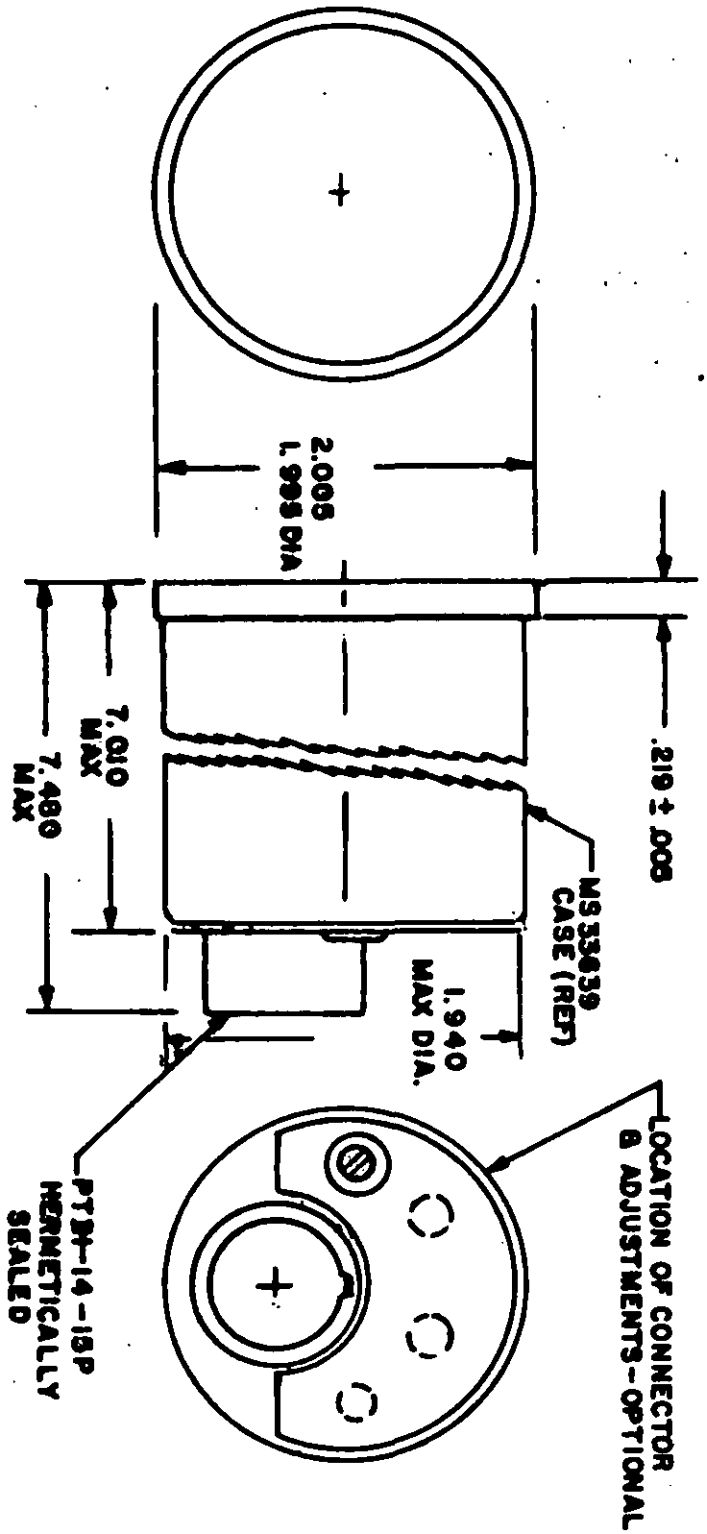
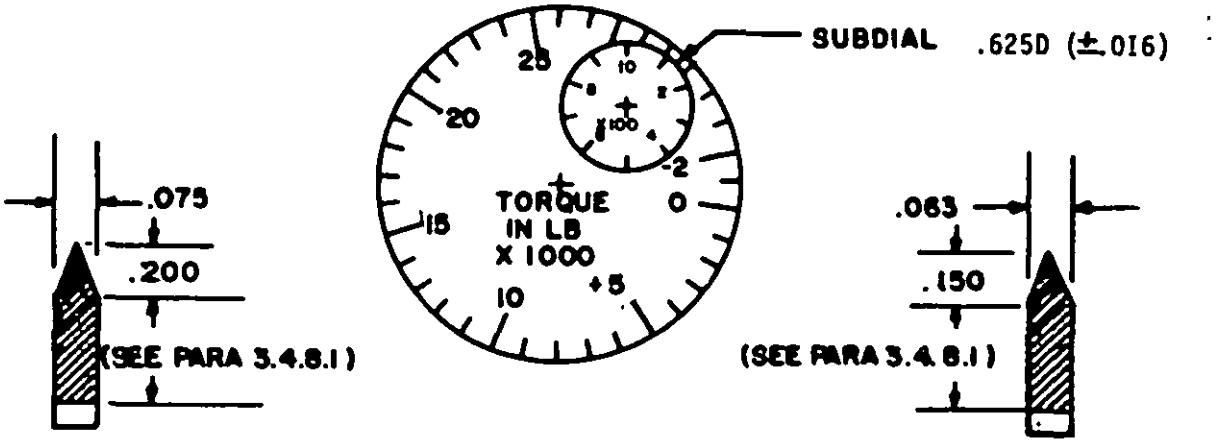


FIG 1. CASE DIMENSIONS

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Main Dial	Height or length Inch ±0.015	Width of line or graduation Inch ±0.005
Numerals: -2, 0, +5, 10, 15, 20, and 25	0.156	0.025
Major graduations	0.187	0.030
Minor graduations	0.125	0.025
Lettering	0.125	0.025
"X 1000"	0.125	0.025
Sub-Dial		
Numerals: 0, 2, 4, 6, 8	0.063	0.025
Major graduations	0.094	0.025
"X 100"	0.046	-----

FIGURE 2. Dial and Pointers.

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white color no. 37875 of FED-STD-595. All markings shall be durable to withstand usage encountered in service. The form of numerals and letters shall be in accordance with MS 33558. Numerals shall distinctly indicate the graduation to which each applies. If practicable, each numeral shall be placed so that the center of area of numeral(s) is on the radial line joining the appropriate graduation and the center of the dial. Any confusion resulting in doubt as to the graduation to which the numeral applies shall be cause for rejection. When several numerals are used in one group, the space between the numerals shall be not less than 1/64 inch.

3.4.7 Dial fasteners. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. If screws are used for attaching the dial, they shall be so located that they will not interfere with dial markings or prevent the markings from being located in the proper position.

3.4.8 Pointer. The pointers of the indicator shall be as light weight as practicable, sufficiently rigid to prevent oscillation under vibration and shall be firmly attached to the mechanism but also shall be readily adjustable. The pointers shall be finished lusterless white, color no. 37875 of FED-STD-595.

3.4.8.1 Pointer length. The tip of the large pointer shall overlap the longest scale graduations by one-third.

3.4.9 External adjustments. The following adjustments and/or switches shall be accessible on the rear of the case: (1) Cal "A" adjustment' single turn potentiometer, range -2900 in. lbs. to -4400 in. lbs., not to exceed a total adjustment range to 2050 in. lbs. (2) Cal "B" adjustment; single turn potentiometer range 24700 in. lbs. to 26800 in. lbs., not to exceed a total adjustment range of 36500 in. lbs. (3) Three position calibration switch (Cal A, B, and R).

3.4.9.1 Switch location. The adjustments and switch shall be permanently attached to the back of the indicator, shall be operable using only a screwdriver, and shall be constructed and mounted on the indicator so as to preclude inadvertent movement of the adjustments or switch position during handling and normal use.

3.4.9.2 Indicator and transmitter matching. The two adjustments and switch shall be sufficient for fitting the indicator to the characteristics of a particular transmitter. It must be possible to match the indicator to the torque transmitter of the engine having the characteristics detailed in para 3.4.10.1.

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3.4.10 Input signal. For flight usage, the input signal shall be obtained from a compatible engine mounted torquemeter pickup assembly, with interconnection between the pickup assembly and the indicator assembly accomplished in accordance with figure 3. The d.c. resistance of either signal source, i.e., either torquemeter pickup coil, is 2.5 ± 0.5 ohms. At 9.2 KHz, its impedance is approximately 4.1 ohms. The characteristics of this signal are as shown in figure 4 and in table I. The indicator shall accept from the torquemeter transmitter a 2-phase signal whose peak-to-peak voltage is constant at 13,820 RPM. The voltage difference between the two phases shall not exceed 10 percent of the larger value and the phase difference shall vary linearly with torque. The relationship that exists in the transmitter between electrical degrees of phase displacement and torque is fixed for any particular transmitter but may vary from transmitter to transmitter in the manner described below:

a. The torque is zero for some phase difference in the interval 48 ±8.4 degrees.

b. The slope of the phase difference versus torque line is 83.3 ±5.0 in-lbs per degree of phase difference.

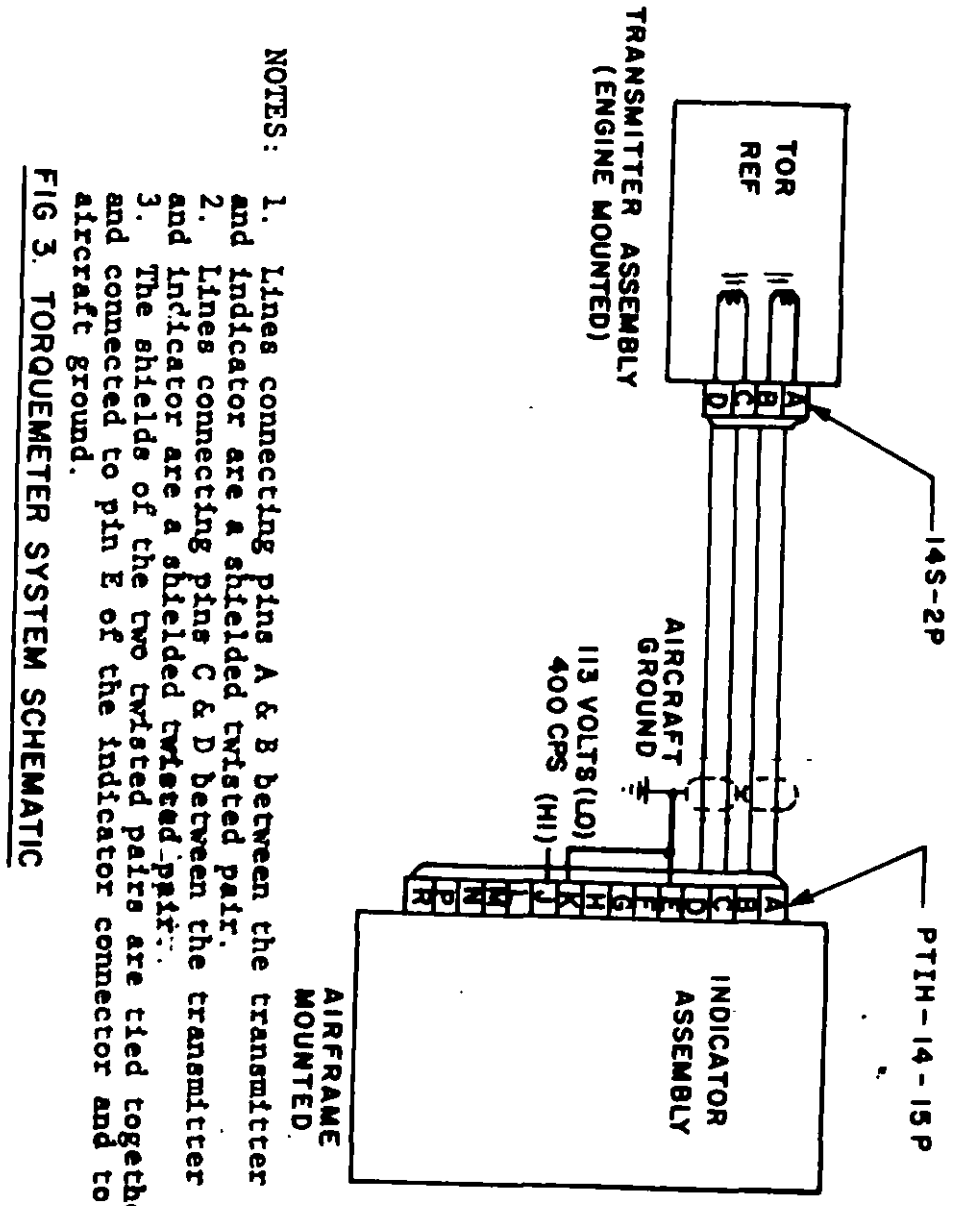
TABLE I

	RANGE		
FREQUENCY, KHz	2.6	4.6	10.0
T, TOTAL PERIOD, MICROSECONDS	385	218	100
X, NEGATIVE SLOPE PORTION OF TOTAL PERIOD, MICROSECONDS	---	100-150	28-38
E, PEAK TO PEAK AMPLITUDE, VOLTS	0.40, MIN 0.60, MAX	0.75, MIN 6.00, MAX	0.75, MIN 6.00, MAX

3.4.11 Output. The angular displacement of the pointers shall be a direct linear and reproducible function of the phase displacement between the input signals. The nominal relationship between indicated torque and input signal phase displacement is approximately 83 in. lbs. per degree of phase displacement.

3.4.12 Dead band. The dead band for the Indicator Assembly shall be defined as that change in phase displacement of the input signals expressed in terms of inch-pounds that can occur without a visible change in main or subdial pointer position. When measured at room ambient and nominal line voltage and frequency, the dead band shall not exceed 68 in. lbs.

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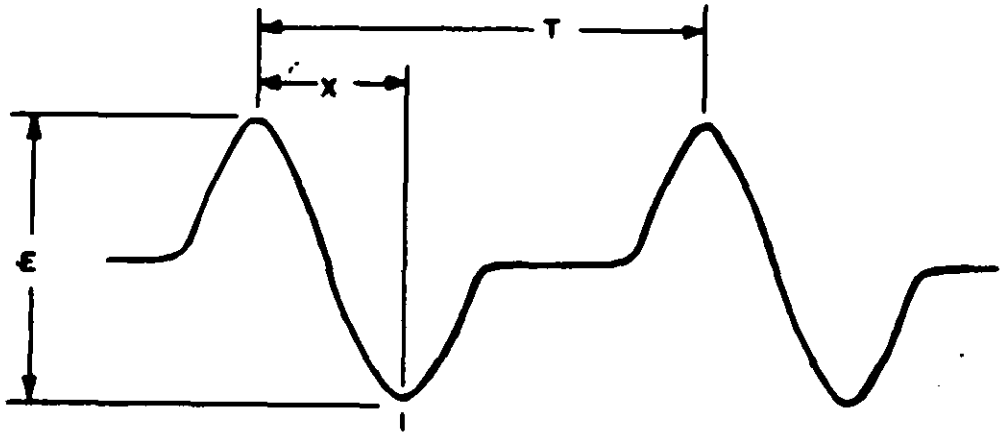


NOTES:
1. Lines connecting pins A & B between the transmitter and indicator are a shielded twisted pair.
2. Lines connecting pins C & D between the transmitter and indicator are a shielded twisted pair.
3. The shields of the two twisted pairs are tied together, and connected to pin E of the indicator connector and to aircraft ground.

FIG 3. TORQUEMETER SYSTEM SCHEMATIC

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Modified Sinusoid



T - Total period

X - Negative Slope portion of total period

E - Peak to Peak Amplitude

FIGURE 4. INPUT SIGNAL WAVEFORM

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3.4.13 Pointer ratio. The gear ratio between the subdial pointer and the main dial pointer shall be 36 to 1 and the subdial pointer shall complete one revolution for each 1,000 in. lbs. change in indication of the main dial pointer.

3.4.14 Frequency response. The frequency response characteristic curve shall be a smooth continuous curve and shall be within the limits shown by the cross-hatched area on figure 5. Frequency response shall be determined when varying the electrical phase displacement of the input signal by 10 degrees or less.

3.4.15 Slewing speed. In response to a step change of 180 degrees input signal phase displacement (equivalent to 14820 in. lbs), the indicator pointers shall be within 160 in. lbs. of the final steady state value within 5 seconds. The indicator shall be designed to have the fastest possible slewing speed consistent with the frequency response requirements of paragraph 3.4.14.

3.4.16 Effect of lead length. With a fixed input signal phase displacement, the indicator reading shall not change by a readable amount when the lead length between indicator and pick-up is changed from 5 to 150 feet of no. 22 AWG wire.

3.4.17 Parallel operation. A maximum of three indicators shall be connected in parallel from one pickup without readable interaction at any pointer position in the calibrated range.

3.4.18 Electrical power. The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when powered from the applicable source with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts. The indicator shall function per the requirements of paragraph 4.2 and subparagraphs when supplied with electrical power as specified below:

Nominal AC Voltage	113 Volts RMS
Nominal AC Frequency	400 Hz
AC Voltage Range	102 to 124 Volts RMS
AC Voltage Variation (Peak to Peak)	±2.5 Volts
AC Frequency Range	360 to 440 Hz
Max. AC Power Required	25 VA

The power factor of the AC power circuits shall be within 0.85 lag to 0.95 lead.

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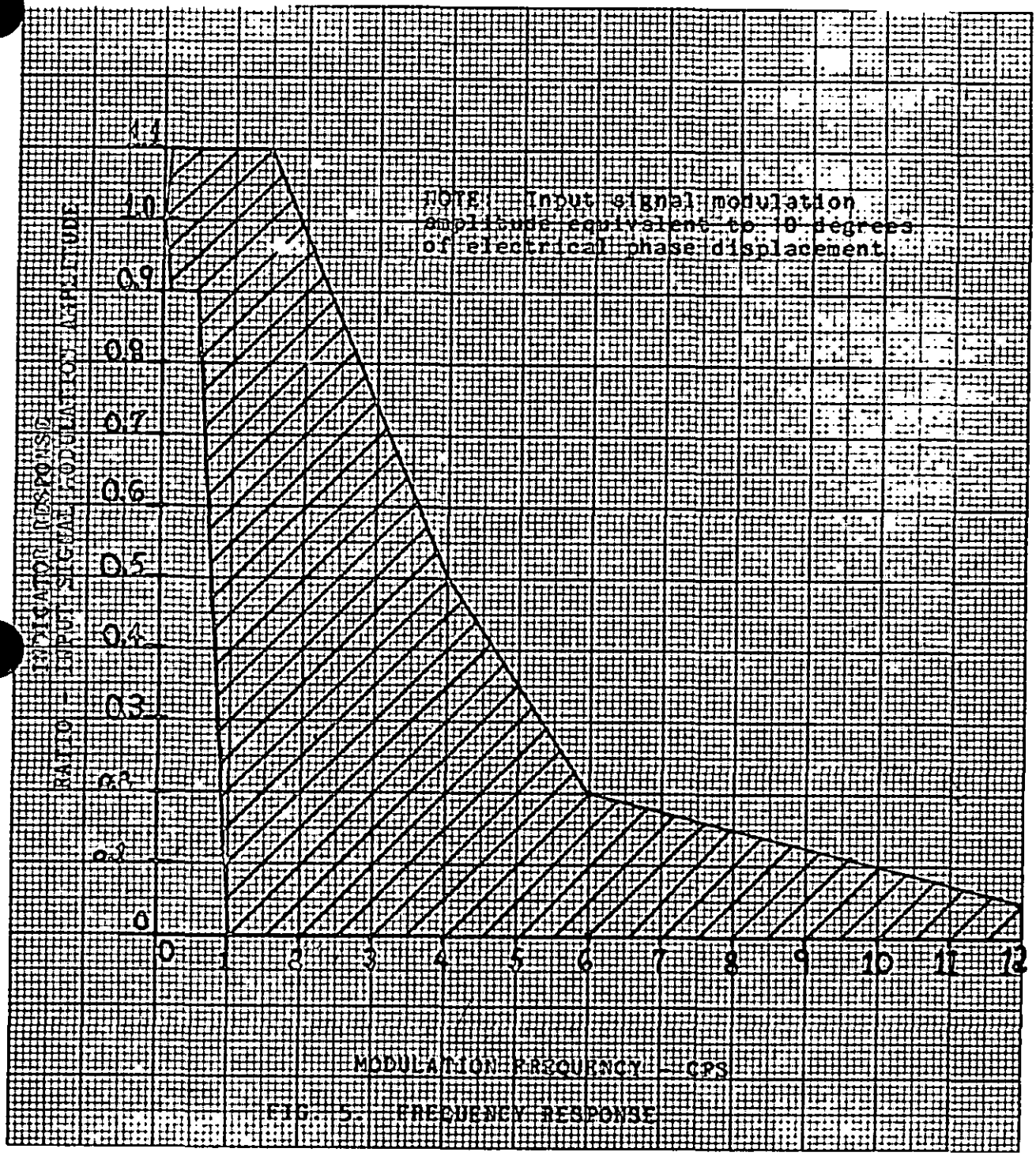


FIG. 5. FREQUENCY RESPONSE

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3.4.18.1 Region B operation. The indicator shall not be damaged by operation in Region B of figure 6. The indicator is not required to function per the requirements of paragraph 4.2 and subparagraphs when operated in Region B. The indicator shall function per the requirements of paragraph 4.2 and subparagraphs in Region A of figure 6.

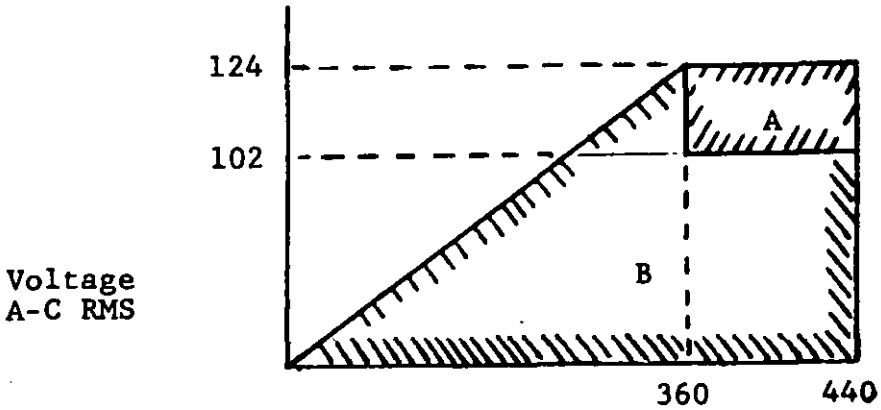


FIG. 6

3.4.19 Power consumption. Power consumption shall not exceed 15 volt amperes at a null or balanced operating condition for the indicator assembly.

3.4.20 Scale error. Scale error shall not exceed the tolerances specified in table II.

3.4.21 Friction and position error. Friction and position error shall not exceed the tolerances specified in table II.

3.5 Performance. The indicator shall operate satisfactorily under the following conditions:

a. Temperature - temperatures ranging from -62° to +71°C.

b. Vibration - vibration in accordance with MIL-STD-810C, method 514.2, category b.1, Procedure I, Table 514/2-II, Figure 514.2-2, curve AR.

c. Altitude - pressure altitudes up to 55,000 feet.

d. Humidity - relative humidity up to 95 percent.

e. Salt fog - exposure to salt fog for a period of 50 hours.

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f. Thermal shock - eight cycles of alternate immersion in tap water at a temperature of 85° ±2°C, and then in tap water at 5° ±2°C (hermetically sealed indicators only).

3.5.1 Reliability program. The contractor shall establish a reliability program in accordance with MIL-STD-785.

3.5.2 Reliability. The indicator shall have a minimum mean-time-between-failures (MTBF) of 2,000 hours based upon a discrimination ratio of 1.5 and a confidence factor of 90 percent.

3.5.3 Dielectric strength. The indicator shall withstand 200V at a commercial frequency applied between pin J or pin K of the electrical connector and the case for a minimum period of 5 seconds.

3.5.4 Voltage and frequency variation. The indicator shall withstand voltage and frequency variations from 102 to 124V and 360 to 440 Hz or any combinations thereof.

3.5.5 Electromagnetic interference. The generation of radio interference by the torquemeter indicator and the vulnerability of the indicator to radio interference shall meet the requirements of MIL-STD-461 for class A1 equipment except that test method CS02 shall apply only to 0.15 MHz.

3.6 Electrical connectors. The type connector used shall be Bendix PN PTIH-14-15P or equivalent.

3.7 Wiring. The indicator shall be wired so that it meets all specification requirements when externally connected as shown in figure 3. Internal wiring shall be color coded, neat and accomplished in such a manner that individual wires may be easily traced.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall goven the manufacturer's part numbers and changes thereto.

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TABLE II. Test tolerances.

Transmitter Position (Degrees)	Indicator Reading (in-lbs)	Tolerances (±in-lbs)			
		Scale Error Vibration Room Temp ($25^{o}C$)	Scale Error High Temp Low Temp High Temp Exposure Low Temp Exposure	Position Error	Friction Error
36	-1000	100	200	50	40
108	+5000	↓	↓	↓	↓
180	+11000	↓	↓	↓	↓
228	+15000	↓	↓	↓	↓
252	+17000	↓	↓	↓	↓
288	+20000	↓	↓	↓	↓
324	+23000	↓	↓	↓	↓

CW rotations from elec. zero shall be positive (+) positions.

CCW rotations from elec. zero shall be negative (-) positions.

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3.9 Jewel bearings. When used, synthetic sapphire or ruby jewel bearings shall be in accordance with MIL-B-2497.

3.10 Finishes and protective coatings.

3.10.1 Aluminum alloy parts. Aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625 except that small holes, straight threads, case inserts, and parts in hermetically sealed enclosures need not be anodized.

3.10.1.1 Exceptions to anodizing. Aluminum alloys that don't anodize satisfactorily shall be coated with a chemical film conforming to MIL-C-5541.

3.10.1.2 Abrasion resistance. When abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.11 Weight. The weight of the indicator shall not exceed 2.50 pounds.

3.12 Screw threads. Unless otherwise specified herein, the threads of all machine screws shall conform to MIL-S-7742.

3.13 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.13.1 Nameplate. A metallic decal type nameplate shall be securely attached to the exterior of the case and shall be marked in accordance with the requirements of MIL-STD-130. The following statement shall appear on the space reserved for special characteristics:

CAUTION - THIS INDICATOR MUST BE CALIBRATED TO THE ENGINE WITH WHICH IT IS USED.

3.14 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

3.14.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the

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performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Qualification Tests (see 4.4).

b. Quality Conformance Tests (see 4.5).

4.3 Inspection conditions & test requirements.

4.3.1 Test conditions. Unless otherwise specified, all tests shall be performed in accordance with the test conditions specified in the applicable test method document or applicable paragraph(s) in this specification.

4.3.2 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, the test shall be conducted at an atmospheric pressure of approximately 29.92 inches Hg and at an ambient temperature of 25° ±5°C.

4.3.3 Master test instrument. The master test instrument is a device designed to simulate the output of the torquemeter sensor throughout the range of the sensor. It must be capable of generating the 2-phase signal described in paragraph 3.4.10 with the phase difference of the given signal known to within ±1/6 degree and with the frequency of the signal known to ±1%.

4.3.4 Signal frequency. When the frequency of the input signal to the indicator is not specifically called out, it shall be 9200 ±300 Hz at an amplitude of 1 to 2 Volts peak-to-peak.

4.4 Qualification testing.

4.4.1 Test samples. The test samples shall consist of 3 indicators representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6, Test methods.

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4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests.

b. Sampling tests.

4.5.1 Individual tests. Each indicator submitted for acceptable shall be subjected to the individual tests. These tests shall determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each indicator accepted shall have passed the following tests:

a. Examination of product.

b. External adjustment.

c. Scale error and friction.

d. Variation in signal frequency.

e. Dielectric strength.

f. Position error.

g. Sealing.

4.5.2 Sampling tests.

4.5.2.1 Sampling plan A tests. Three indicators shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests.

b. Low temperature.

c. High temperature.

d. Fogging.

e. Vibration.

f. Magnetic effect.

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4.5.2.2 Sampling plan B tests. Unless otherwise specified, 3 indicators shall be selected at random from the first 15 on the contract and subjected to the following tests as described under 4.6.

a. Sampling plan A tests.

b. Power consumption.

c. Voltage and frequency variation.

d. Temperature - Altitude.

e. Low temperature exposure.

f. High temperature exposure.

g. Thermal shock.

h. Salt fog.

4.5.2.3 Sampling plan C tests. Samples shall be selected in accordance with MIL-STD-105 to yield acceptable quality level (AQL) of 2.5 percent defective, using the monthly production lot as a basis for sample selection, not to exceed a maximum of 50 units per sample. These units shall be subjected to the following tests as described under 4.6:

a. Individual tests.

b. Low temperature.

c. High temperature.

d. Vibration.

e. Fogging.

4.5.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of the failure have been determined and appropriately corrected.

4.5.4 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

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4.5.5 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5.6 Disposition of samples upon completion of test. Any sample tested may be delivered on contract provided it meets all the following requirements:

a. It is representative of production units currently being accepted.

b. It is in "good as new" condition, or has been refurbished (see para 6.4).

c. It is otherwise satisfactory.

4.6 Test methods.

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and markings. Prior to assembly, dials, pointers, and cover glass shall be inspected as to markings, dimensions, and finishes.

4.6.2 Scale error and friction. The indicator shall be connected to the test instrument and the system energized. The indicator shall then be subjected to a scale error and friction test at room temperature at the test points listed in table II. The friction error is the difference in readings made before and after tapping, and shall not exceed 40 in-lb at any test point.

4.6.3 Position error. The indicator pointer shall be positioned at the electrical zero reference mark with power applied. The change in pointer position when the indicator is held in any position shall not exceed the value specified in table II.

4.6.4 Vibration. The indicator shall be subjected to a vibration test in accordance with MIL-STD-810C, Method 514.2, equipment category b.1, Procedure I, Table 514.2-II, Figure 514.2-2, Curve AR. Upon completion of this test, the indicator shall be subjected to the scale error test at room temperature. The scale errors shall not exceed the tolerances specifiec in table II.

4.6.5 Low temperature. The indicator shall be subjected to a temperature of $-54^{\circ} \pm 2^{\circ}C$ for a period of 4 hours. While at this temperature, it shall be checked for scale error and friction. The tolerances shall not exceed those specified in table II.

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4.6.6 High temperature. The indicator shall be subjected to a temperature of 71°C for a period of 4 hours. While at this temperature, it shall be checked for scale errors. The scale error and friction shall not exceed the tolerances specified in table II.

4.6.7 Variation in signal frequency. The scale error test shall be repeated at signal frequencies of 6700 ±100 Hz, 10,000 ±100 Hz, and 2700 ±100 Hz. At the 2700 ±100 Hz frequency, the signal amplitudes shall be $0.40 \pm {}^{0.04}_{0}$ Volts peak-to-peak. The scale error at 6700 Hz and 10,000 Hz shall not exceed 115 in-lb. At 2700 Hz, the scale error shall not exceed 230 in-lb and there shall be no oscillation or erratic movement of the pointer.

4.6.8 Dielectric strength. A potential of 200V at a commercial frequency shall be applied between pin J and pin K of the electrical receptacle and the case of the indicator for a minimum period of 5 seconds. There shall be no breakdown of the insulation.

4.6.9 External adjustment.

4.6.9.1 Input power supply. Input power supply shall be adjusted to 113 ±2 Volts at 400 ±5 Hz.

4.6.9.2 Cal A adjustment. The indicator selector switch shall be placed in the "Cal A" position. Using the Cal A potentiometer:

a. Indicator shall be checked for range of adjustment from -2900 to -4400 in. lbs.

b. Range of adjustment shall not exceed 2050 in. lbs.

c. Clockwise rotation of adjustment potentiometer shall cause clockwise rotation of the main pointer.

d. Cal A potentiometer shall be adjusted to produce a -4000 in. lbs. indication.

4.6.9.3 Cal B adjustment. The indicator selector switch shall be placed in the Cal B position. Using the Cal B potentiometer:

a. Indicator shall be checked for range of adjustment from 24,700 to 26,800 in. lbs.

b. Range of adjustment shall not exceed 3650 in. lbs.

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c. Clockwise rotation of the adjustment potentiometer shall cause clockwise rotation of the main pointer.

d. Cal B potentiometer shall be adjusted to produce a 26,000 in. lb. indication.

4.6.10 Voltage-frequency variation. The accuracy of the indicator shall be ±114 in. lbs., when the input signal is varied from 36 to 324 electrical degrees with the signal frequency at 9200 ±300 Hz at an amplitude of 1 to 2 Volts peak-to-peak at the following power supply combinations:

102 Volts ±2 Volts	360 ±5 Hz
102 Volts ±2 Volts	440 ±5 Hz
115 Volts ±2 Volts	400 ±5 Hz
124 Volts ±2 Volts	360 ±5 Hz
124 Volts ±2 Volts	440 ±5 Hz

4.6.10.1 Linearity. A linear relationship of ±114 in. lbs. of indicator reading to input signal phase displacement shall be demonstrated throughout the range of 36 to 324 electrical degrees.

4.6.11 Sealing. The hermetically sealed indicator shall be tested for leaks at room temperature by means of a mass-spectrometer-type helium leak detector. The detected leak rate for the 10-percent helium and 90-percent nitrogen filling medium shall not exceed 0.1 micron cubic foot per hour and for the 100-percent helium filling medium shall not exceed 1.0 micron cubic foot per hour at a pressure differential of 12 to 14.7 psi.

4.6.12 Power consumption. The indicator shall be properly connected and power shall be applied for a period of five minutes. The power consumption shall be measured at the end of this time and shall not exceed 15 Volt Amperes. The power factor of the a.c. power circuits shall be within 0.85 lagging to 0.95 leading.

4.6.13 Magnetic effect. The indicator, first, not operating and second, operating on rated voltage, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. Deflection of the compass at any of the specified positions shall not exceed 3°.

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4.6.14 Effect of lead length. The indicator shall not change more than 1° of arc when the lead length between the indicator and transmitter is changed from 5 to 500 feet of no. 20 AWG gauge wire. Equivalent resistances may be substituted for the added wire length. The ground lead shall be kept short.

4.6.15 Temperature-altitude. The temperature-altitude test shall be conducted in accordance with MIL-STD-810, method 504.1, procedure I (except the temperature shall be -54°C) with a chamber pressure equivalent to an altitude of 55,000 feet. The rate of pressure change shall exceed an average of 3 psi per second while under these conditions and at the end of a 1-hour period, the scale error and friction test shall be conducted. The errors shall not exceed the tolerances specified in table II. This test may be conducted concurrently with the low temperature exposure test.

4.6.16 Low temperature exposure. The indicator shall be subjected to a temperature of -62° ±2°C for a period of 48 hours. The temperature shall then be changed to -54° ±2°C and maintained for an additional 24 hours. The indicator shall then be tested at this temperature for scale and friction errors. The errors shall not exceed those specified in table II. The indicator shall then be returned to room temperature for a period of not less than 4 hours and again subjected to the scale and friction error tests of table II. The indicator shall not exceed the scale and friction error specified in table II. There shall be no damage or deterioration that will affect subsequent operations.

4.6.17 High temperature exposure. The indicator shall be subjected to a temperature of 71°C for a period of 24 hours and then tested at this temperature for scale errors and friction as specified in table II. The scale error and tolerances and friction shall not exceed the tolerances specified in table II. There shall be no damage or deterioration that might affect subsequent operation.

4.6.18 Fogging. The indicator shall be placed in a controlled ambient temperature of 85° ±5°C for a minimum period of 2 hours. Within 10 seconds after removal from the oven, the indicator shall be immersed to a depth of 1/4 inch in an ice water bath maintained at 0° to 7°C. After 3 minutes, the indicator shall be removed from the fluid bath and quickly wiped dry (compressed air shall not be used). The cover glass and dial shall immediately be inspected and any evidence of fogging or damage shall be cause for rejection.

4.6.19 Thermal shock. The indicator shall be subjected to the temperature shock test in accordance with MIL-STD-810,

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method 503.1, procedure I. At the completion of this test, the indicator shall be subjected to the scale and friction error tests of table II. The indicator shall not exceed the tolerances for room temperature in table II.

4.6.20 Salt fog. The indicator shall be subjected to a salt fog test for a period of 48 hours in accordance with MIL-STD-810, method 509.1, procedure I. There shall be no evidence of external corrosion or deterioration that might affect subsequent operation. The scale errors after this test shall not exceed the tolerances specified for room temperature scale error in table II.

4.6.21 Electromagnetic interference. The indicator shall be tested for conducted and radiated radio noise interference in accordance with para 3.5.5 herein, and MIL-STD-462.

4.6.22 Engine compatibility test. The qualification inspection sample units shall be connected to the torquemeter assembly on the engines of the aircraft and sufficient calibration checks and engine run-up conducted to demonstrate compatibility of the indicators with the torquemeter assembly. Any incompatibility shall be cause for rejection. The facilities for performing this test will be furnished by the procuring agency.

4.6.23 Reliability. Seven indicators shall be tested for reliability. Acceptance permitting production and total test time shall be in accordance with MIL-STD-781, test plan I. Recording, data handling, and reporting shall also be in accordance with MIL-STD-781.

4.6.23.1 Test. Qualification Phase and Production Acceptance (Sampling) Phase Tests of MIL-STD-781 shall be conducted. Test level E of MIL-STD-781 shall be used. The indicators shall be energized during the heating portion of the temperature cycle and supplied with a signal to cause it to be cycled as indicated in table III. Upon completion of each 125 hours of cycling, the indicators shall be checked for scale error and friction. The errors shall not exceed the tolerances specified in table II. If an indicator does not meet these tolerances, it shall be counted as a failure. The contractor shall prepare a report in accordance with MIL-STD-781 which shall contain test procedures, details of any malfunctions and corrective actions taken.

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TABLE III

Approx Value of Applied Signal in In - lbs	Duration of Signal - Minutes
- 1,000	30
+ 5,000	30
+11,000	30
+17,000	30
+23,000	30
- 1,000	20
POWER OFF	10

4.7 Inspection and testing of preparation for delivery. Preparation for delivery shall be tested in accordance with MIL-P-116 to insure compliance with section 5 of this specification.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be in accordance with 5.1.1 or 5.1.2, as specified (see 6.2).

5.1.1 Level A (reference definition in MIL-STD-794). Indicators shall be preserved and packaged in accordance with MIL-P-116, method 1C-1, one each in a container conforming to PPP-B-636, weather-resistant class. Cushioning shall be of a type, density, configuration, and thickness that will insure that shock transmission will not exceed the peak shock g-value specified for transmitters when completed packs are submitted to the rough-handling drop tests of MIL-P-116.

5.1.2 Level C. Indicators shall be individually preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration, or physical damage during shipment from the supply source to the first receiving activity. This level may conform to the supplier's standard practice provided the latter meets the requirements of this level.

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5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. Shipments that exceed the weight limitations of this specification shall be packed in overseas-type containers conforming to PPP-B-601. The gross weight of each PPP-B-601 container shall not exceed 200 pounds. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636 or PPP-B-601, as applicable. Waterproofing shall be in accordance with MIL-STD-1186.

5.2.2 Level B. Indicators preserved and packaged as specified in 5.1.1 shall be packed in domestic shipping containers conforming to PPP-B-636. Where practicable, exterior shipping containers shall be of uniform shape and size and of minimum cube and tare. The gross weight of each shipping container shall not exceed the weight limitations of the specification. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Physical protection. Cushioning, blocking, bracing, and waterproofing shall be in accordance with MIL-STD-1186. Waterproofing requirements for cushioning materials and containers shall be waived for domestic shipments. The drop tests of MIL-STD-1186 shall be waived when the item is preserved, packaged, and packed for immediate use or when the drop tests of MIL-P-116 are applicable.

5.4 Marking. In addition to any other marking required by the order or contract, interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

INDICATOR, TORQUEMETER, TYPE AEU-13/A

5.4.1 Reinspection date markings. Reinspection date markings shall be in accordance with the instructions of the procuring activity.

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6. NOTES

6.1 Intended use. The indicators covered by this specification are intended for use in aircraft powered with turboprop engines equipped with a torquemeter transmitter whose output signal is as described in paragraph 3.4.11.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Information, if any, to be submitted with qualification samples.

c. When sampling plan B tests will not be conducted.

d. Levels of packaging and packing required and reinspection date markings (see section 5).

e. Option of either paragraph 4.5.2.1, Sampling plan A tests, or paragraph 4.5.2.3, Sampling plan C tests, applied to each procurement or contract if Sampling plan B test is not to be conducted.

6.3 Drawing requirements. The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of MIL-STD-100.

6.4 Definitions.

6.4.1 Amplitude. Whenever the work "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.4.2 Hermetic seal. A hermetic seal shall be a perfectly closed and air tight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.4.3 Good as new. "Good as new" shall mean indicators operated less than 10 percent of the specified MTBF operation.

6.4.4 Refurbished. Refurbished shall mean that the indicator has been completely overhauled with all component parts meeting current parts standards; and the indicator shall have been subjected to and met all the requirements of a new indicator.

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6.4.5 Corrosion resistance. The terms "corrosion resistant" and "protected as specified herein to resist corrosion" as used in this specification, shall mean a material which exhibits no visual evidence of corrosion subsequent to exposure to the Salt Fog Test. The material shall have been cleaned by ultrasonic methods or equivalent prior to testing to determine corrosion resistance.

6.5 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

6.6 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Hq AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016. Information pertaining to qualification of products to this specification may be obtained from that activity.

Custodians:
Navy - AS
Air Force - 99

Preparing Activity:
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